

Notation. Let $\alpha: I \rightarrow \mathbb{R}^3; t \mapsto (\alpha_1(t), \alpha_2(t), \alpha_3(t))$ be a regular Curve.

$$\frac{d\alpha}{dt} = \frac{d}{dt} \alpha(t) = \left(\frac{d}{dt} \alpha_1(t), \frac{d}{dt} \alpha_2(t), \frac{d}{dt} \alpha_3(t) \right).$$

Re Parametrization

Given a regular Curve $\alpha: I \rightarrow \mathbb{R}^3$, generally, $|\frac{d\alpha}{dt}| \neq 1$.

But, Consider

$$s = s(t) = \int_{t_0}^t \left| \frac{d\alpha}{du} \right| du. \quad \text{Then, } \frac{ds}{dt} = \left| \frac{d\alpha}{dt} \right|$$

$$\text{Now, } \frac{d\alpha}{ds} = \frac{d\alpha}{dt} \cdot \frac{dt}{ds} = \frac{d\alpha}{dt} \cdot \left| \frac{d\alpha}{dt} \right|^{-1}.$$

Def. Let $\alpha: I \rightarrow \mathbb{R}^3$ be a unit speed curve.

The tangent vector $T(s) = \alpha'(s)$

The normal vector $N(s) = \alpha''(s)/K(s)$ where $K(s) = \|\alpha''(s)\|$.

The binormal vector $B(s) = T \times N$

The torsion $\tau(s) = -\langle B'(s), N(s) \rangle$

$$\begin{pmatrix} T' \\ N' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & K & 0 \\ -K & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

Prop. Let $\alpha(s)$ be a unit speed curve with $k \neq 0$. The following are equivalent:

- a). α is a Plane Curve.
- b). The torsion is identically zero.

Proof. WLOG, suppose that α is a plane curve on the plane $z=0$.

Then, we can write $\alpha(s) = (x(s), y(s), 0)$, for some differentiable x, y .

Then, the tangent and normal are

$$T(s) = (x'(s), y'(s), 0) \text{ and } N(s) = (-y'(s), x'(s), 0).$$

So, the binormal is

$$B = T \times N = (0, 0, x'^2 + y'^2) = (0, 0, 1).$$

Now, the B -normal is constant, so $B' = (0, 0, 0)$, so $\tau = \langle B', N \rangle = 0$.

Conversely, suppose that the torsion is zero. Then, since $B' = -\tau N = 0$,

so the B -normal vector B is constant. Now,

$$\frac{d}{ds} \langle \alpha(s) - \alpha(0), B \rangle' = \langle \alpha'(s), B \rangle + \langle \alpha(s) - \alpha(0), B' \rangle = \langle \alpha'(s), B \rangle = 0.$$

Therefore, $\langle \alpha(s) - \alpha(0), B \rangle$ is constant, but $s=0$, $\langle \alpha(s) - \alpha(0), B \rangle = 0$,
so $\langle \alpha(s) - \alpha(0), B \rangle \equiv 0$. Now, $\alpha(s)$ lies in a plane containing $\alpha(0)$
perpendicular to B . $\square$